

AI-Powered Customer Support & Service Automation

Project Documentation

Submitted By

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Project Title: AI-Powered Customer Support & Service Automation

Platform : Salesforce Service Cloud

Technologies Used :

- Salesforce Service Cloud
 - Flow Builder
 - Prompt Builder
 - Agentforce
 - Validation Rules
 - Queues
 - Reports
 - Dashboards
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1. Introduction

Customer support plays an important role in every business. Managing customer complaints manually often results in delayed responses, incorrect assignments, and inconsistent service quality.

This project introduces an AI-powered customer support system built using Salesforce Service Cloud. The application automates customer case handling through intelligent routing, validation rules, AI-generated summaries, reports, and dashboards, making customer service faster, smarter, and more efficient.

2. Problem Statement

Traditional customer support systems face several challenges:

- Manual case assignment
- Delayed response time
- Human errors
- Difficulty in tracking cases
- Lack of automation
- No AI assistance for support agents

These issues reduce customer satisfaction and increase workload for support teams.

3. Proposed Solution

The proposed solution is an AI-powered customer support application developed using Salesforce Service Cloud.

The system automatically validates customer cases, assigns them to the appropriate support queue based on issue category, and generates AI-powered summaries to assist support agents. Reports and dashboards provide real-time insights into customer support activities.

4. Objectives

The objectives of this project are:

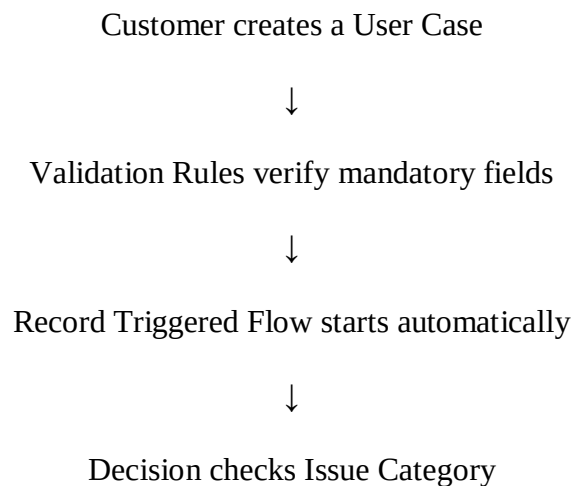
- Automate customer support operations
 - Reduce manual work
 - Improve response time
 - Route cases automatically
 - Generate AI summaries
 - Monitor cases using dashboards
 - Improve customer satisfaction
-

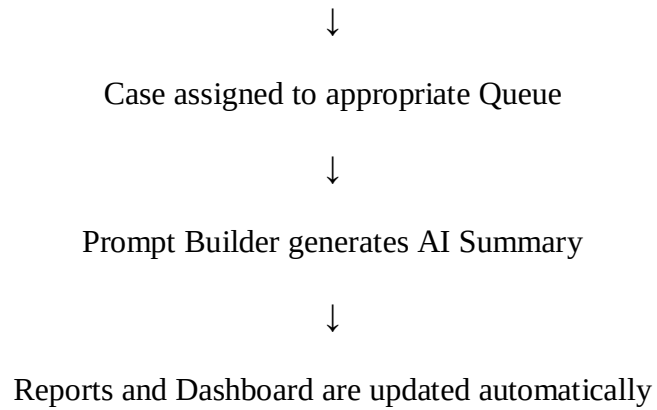
5. Technologies Used

Technology	Purpose
Salesforce Service Cloud	CRM Platform
Flow Builder	Automation
Prompt Builder	AI Summary Generation
Validation Rules	Data Validation
Queues	Automatic Case Assignment
Reports	Data Analysis
Dashboards	Visualization

6. System Architecture

The workflow of the application is:





7. Project Features

The application includes the following features:

- User Case Management
- AI Generated Summary
- Automatic Queue Assignment
- Validation Rules
- Record Triggered Flow
- Prompt Builder
- Reports
- Dashboard
- Real-Time Monitoring

8. Project Implementation

8.1 Lightning App

A dedicated Lightning Application named **AI Customer Support Automation** was created to manage customer support activities.

The application includes:

- Customers
 - Products
 - Orders
 - User Cases
 - Reports
 - Dashboards
 - Knowledge
-

8.2 User Case Object

A custom object named **User Case** was created with fields such as:

- User Case Name
- Product
- Order
- Priority
- Issue Category
- AI Summary

This object stores all customer support requests.

9. Validation Rules

Validation Rules ensure data quality before saving records.

Implemented Rules:

Damage Cases - Priority is mandatory.

Warranty Cases - Priority is mandatory.

These rules prevent incomplete records from being saved.

10. Queue Management

Support queues were created to distribute customer cases automatically.

Damage Support Queue - Handles all damage-related customer issues.

Warranty Support Queue - Handles warranty-related customer issues.

11. Record Triggered Flow

A Record Triggered Flow automates customer case assignment.

Workflow:

User Case Created



Decision Element checks Issue Category



Damage



Assign Damage Queue

OR

Warranty



Assign Warranty Queue

The process executes automatically without manual intervention.

12. Prompt Builder

Prompt Builder is used to generate AI-powered customer support summaries.

The prompt generates:

- Issue Summary
- Priority Analysis
- Recommended Action

This helps support agents understand customer issues quickly.

13. Reports

The project includes reports for monitoring customer cases.

Examples:

- User Cases Report
- User Cases by Issue Category
- User Cases by Priority

These reports help analyse customer support data efficiently.

14. Dashboard

A Customer Support Dashboard was created.

Dashboard Components include:

- User Cases Table
- User Cases by Issue Category
- User Cases by Priority

The dashboard provides a visual overview of support activities.

15. Testing

The following scenarios were tested successfully.

Test Case	Result
Create User Case	Passed
Validation Rule	Passed
Queue Assignment	Passed
Flow Execution	Passed
Prompt Builder	Passed
Report Generation	Passed
Dashboard Update	Passed

16. Results

The developed application successfully:

- Reduced manual effort
- Improved case routing

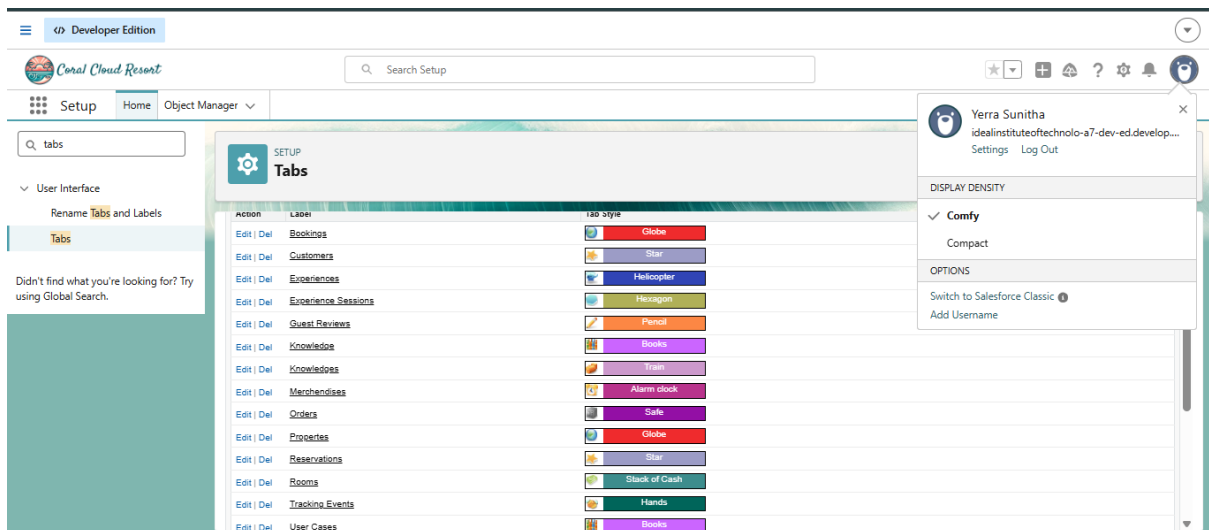
- Generated AI summaries
- Automated support workflows
- Improved reporting
- Increased operational efficiency

17. Future Enhancements

Future improvements may include:

- Email Notifications
- Chatbot Integration
- Omni-Channel Support
- AI Case Prediction
- Sentiment Analysis
- Customer Self-Service Portal
- Mobile Support
- Einstein AI Recommendations

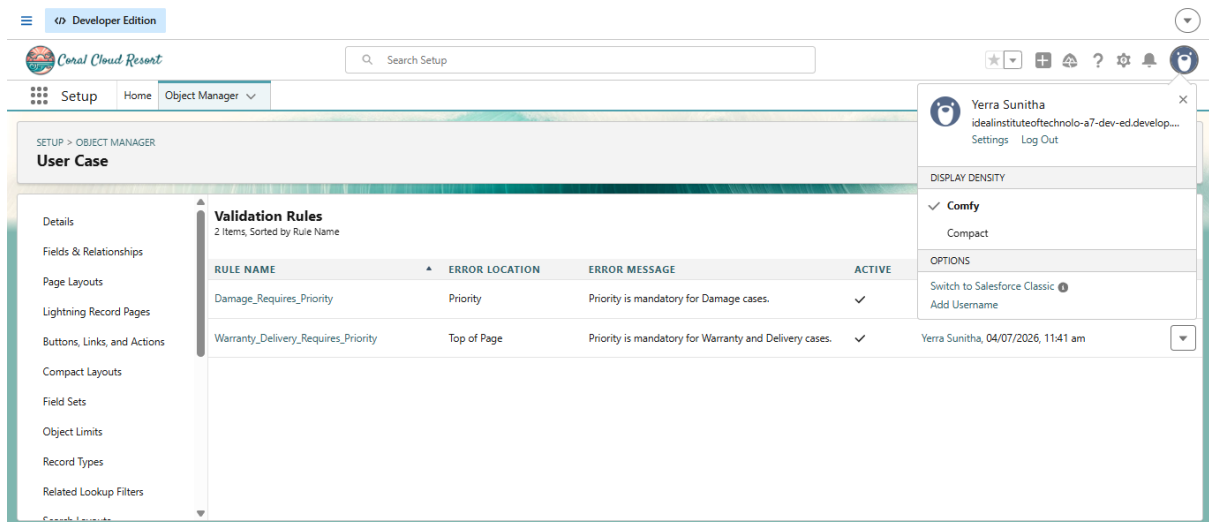
18. Outputs



Tabs Configuration

Description:

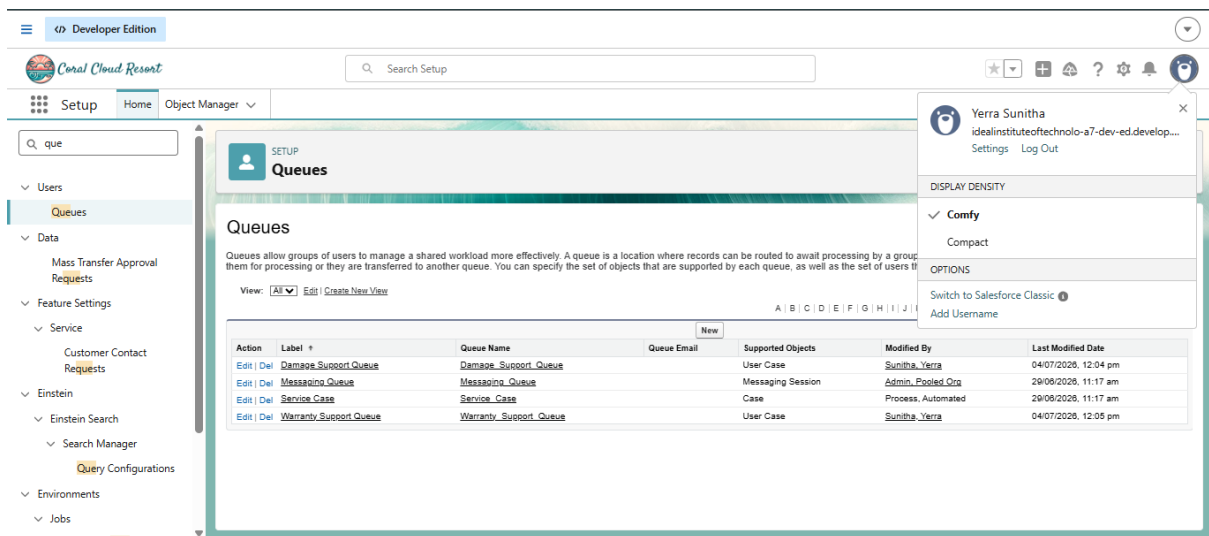
Configuration of custom tabs used in the AI Customer Support Automation application, enabling easy navigation to User Cases, Reports, Dashboards and related components.



User Case Validation Rules

Description:

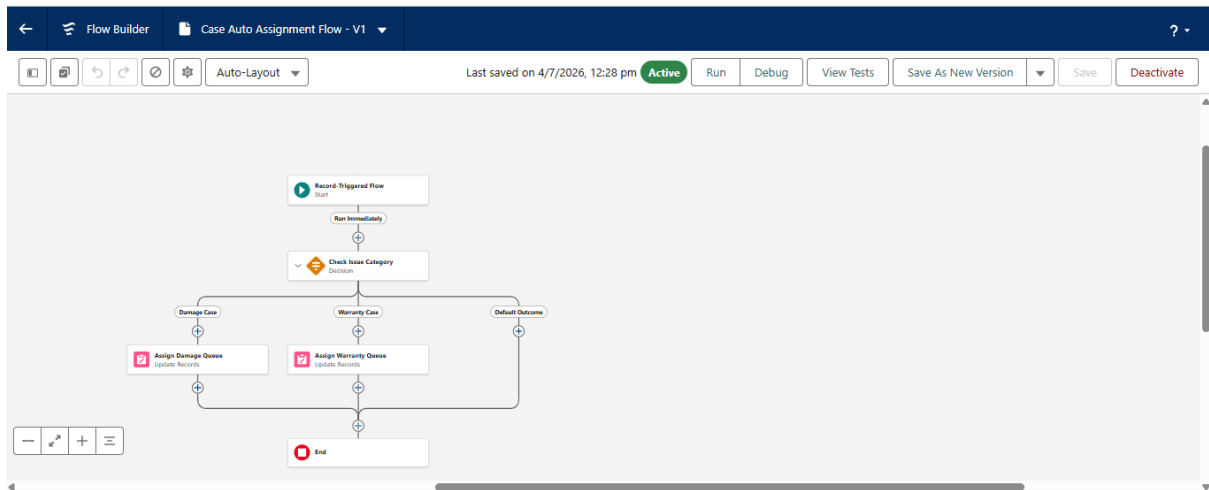
Validation rules implemented to ensure mandatory Priority values for Damage and Warranty issue categories before saving User Case records.



Queues Configuration

Description:

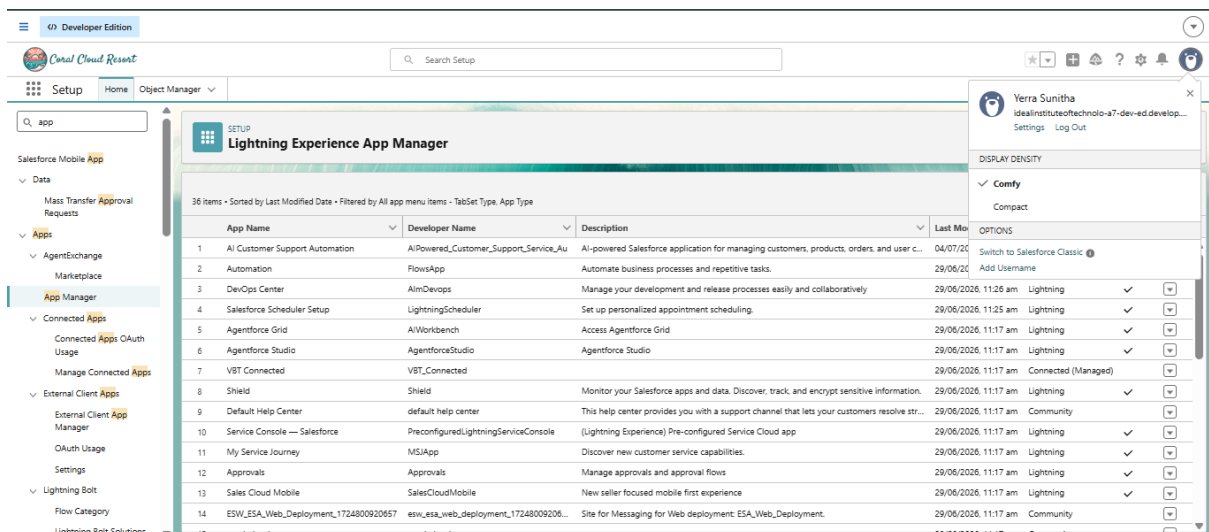
Configuration of Salesforce Queues used for automatic routing of customer support cases based on issue category.



Case Auto Assignment Flow

Description:

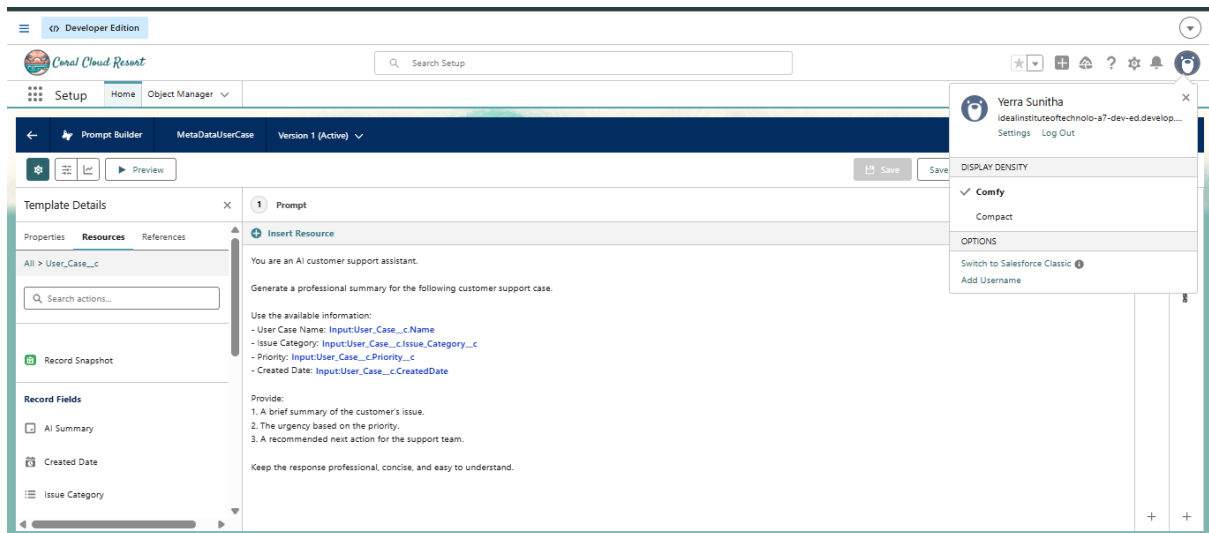
Record-Triggered Flow developed using Flow Builder to automatically assign User Cases to the appropriate support queue based on the selected Issue Category.



Lightning Experience App Manager

Description:

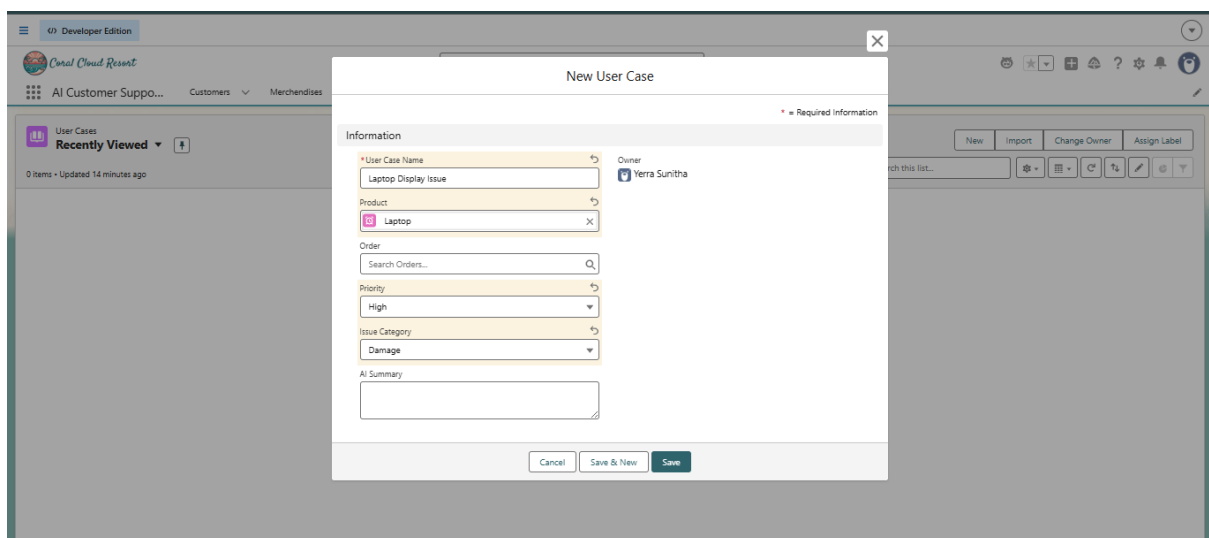
Custom Lightning Application named **AI Customer Support Automation** configured in Salesforce App Manager to provide a centralized workspace for customer support operations.



AI Prompt Builder

Description:

Prompt Builder configuration used to generate AI-powered summaries for User Cases using Salesforce AI capabilities.



New User Case Form

Description:

User interface for creating a new User Case with details such as Product, Order, Priority, Issue Category, and AI Summary.

Developer Edition

Coral Cloud Resort

Search Setup

Setup

Home

Object Manager

App Launcher

Q ai

Lightning Experience App Manager

New Lightning App

New External Client App

Apps

AI Customer Support Automation

Items

Approval Submission Details

Campaigns

Capability Details

Email Templates

Field Audit Trail

View More

Sales Cloud Everywhere

ADMINISTRATION

Users

Data

Email

PLATFORM TOOLS

Subscription Management

Sorted by App Name • Filtered by All app menu items - TabSet Type, App Type

App Name	Developer Name	Description	Last Modified ...	App T...	Vi...
Agentforce Grid	AIWorkbench	Access Agentforce Grid	29/06/2026, 11:17 am	Lightning	✓
Agentforce Studio	AgentforceStudio	Agentforce Studio	29/06/2026, 11:17 am	Lightning	✓
Customer Support Automati...	AIPowered_Customer_Support_Service_Au	AI-powered Salesforce application for managing customers, products, or...	06/07/2026, 2:13 pm	Lightning	✓
AI Tabs	AITabSet		29/06/2026, 11:17 am	Classic	✓
Analytics Studio	Insights	Build CRM Analytics dashboards and apps	29/06/2026, 11:17 am	Classic	✓
6 App Launcher	AppLauncher	App Launcher tabs	29/06/2026, 11:17 am	Classic	✓
7 Approvals	Approvals	Manage approvals and approval flows	29/06/2026, 11:17 am	Lightning	✓
8 Automation	FlowsApp	Automate business processes and repetitive tasks.	29/06/2026, 11:26 am	Lightning	✓
9 Bolt Solutions	LightningBolt	Discover and manage business solutions designed for your industry.	29/06/2026, 11:17 am	Lightning	✓
10 Community	Community	Salesforce CRM Communities	29/06/2026, 11:17 am	Classic	✓
11 Content	Content	Salesforce CRM Content	29/06/2026, 11:17 am	Classic	✓
12 Coral Cloud Resorts	Coral_Cloud	Your smart hospitality app.	29/06/2026, 11:17 am	Lightning	✓

Developer Edition

Coral Cloud Resort

Search...

AI Customer Suppo...

Customers

Merchandises

Orders

User Cases

Knowledge

Dashboards

Customers

Recently Viewed

New

Import

Change Owner

Assign Label

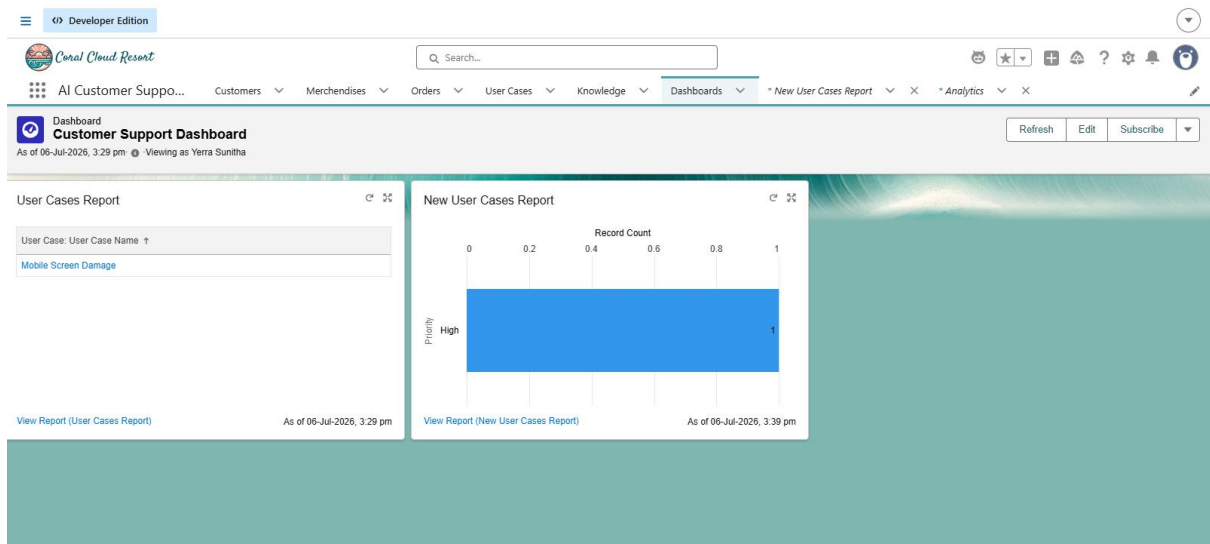
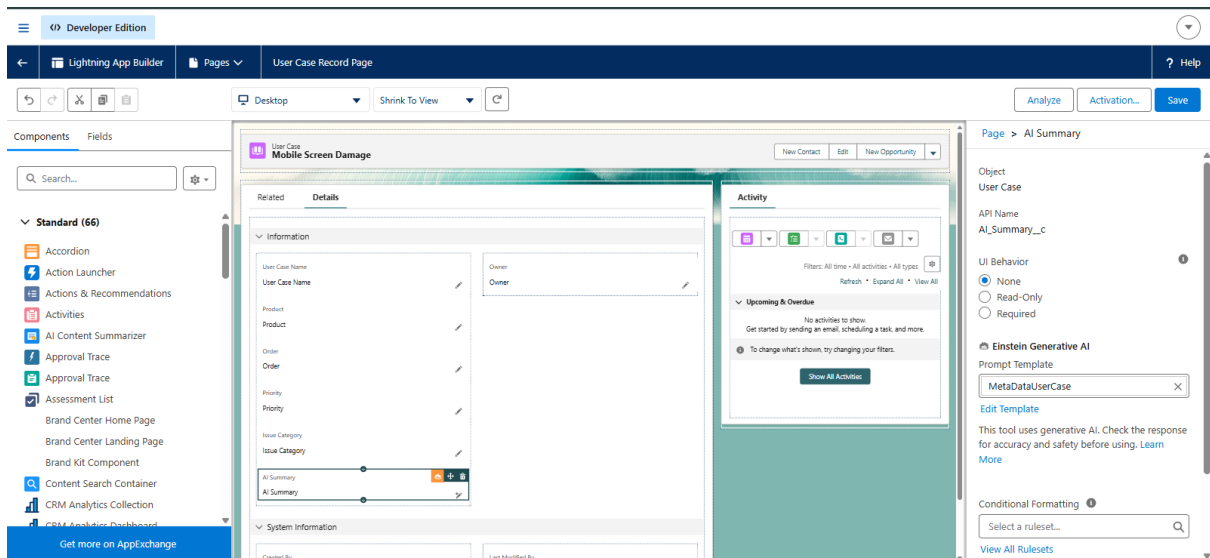
1 item • Updated an hour ago

Search this list...

Customer Name

1

Text



Customer Support Dashboard

Description:

The dashboard provides a centralized view of User Cases using reports and charts, enabling administrators to monitor support activities efficiently.

19. Conclusion

The AI-Powered Customer Support & Service Automation project demonstrates how Salesforce Service Cloud can be used to automate customer service operations efficiently.

By integrating Validation Rules, Record Triggered Flows, Queues, Prompt Builder, Reports, and Dashboards, the application reduces manual effort, improves productivity, and enhances customer satisfaction. The project provides a scalable foundation for future AI-driven customer support solutions.

✦ Acknowledgement

I would like to express my sincere gratitude to my mentors, faculty members and Salesforce Trailhead for providing the knowledge and resources required to complete this project successfully. I also thank everyone who supported and guided me throughout the development process.